Whose Contribution Counts?

AI Mediation and Collective Authorship in Digital Improvisation

Roman Jahandideh

Exploratory proposal for discussion with Dr. Rebecca Caines
Graduate Program in Communication & Culture, York University
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Working proposal prepared for discussion with Dr. Rebecca Caines. The research setting and project partnerships will be developed in consultation with the prospective supervisor.

1 Problem and cultural stakes

In collaborative creative tools, a generative model does more than produce content: it *reinterprets* what people contribute. Someone types a phrase, and a moment later it has become a sound or a moving form in a shared scene. The group then has to decide what that transformation means, and whether it still belongs to the person who offered it.

When AI translates creative contributions into shared audiovisual material, its outputs may become more visible or influential than the contributions that prompted them, a possibility to investigate rather than an established effect. This raises a cultural question about whose interpretations gain authority, whose contributions remain recognisable, and how disagreement is negotiated in collective creative practice. Creative platforms increasingly record who supplied what; this project asks whether such records settle questions of credit.

Improvisation is a good place to ask. Improvisers already have conventions for listening to, building on and refusing one another's offers, and improvisation has been argued to be a research method grounded in risk, active listening and collaborative response (Caines, 2020). A system that transforms and labels offers enters those conventions, and may quietly change them.

2 Related work

Existing research examines human–AI co-creation, collective decision-making and the attribution of creative credit; this project investigates their intersection in small-group digital improvisation. Table 1 compares the closest studies.

Several elements already exist: group negotiation around a shared AI, with "unacknowledged contributions" coded as a breakdown (Xu et al., 2026); AI reshaping social roles in co-creation (Suh et al., 2021); and improvisation informing AI partners and performers (Trajkova et al., 2024; Mathewson and Mirowski, 2017).

Table 1: Closest related work and the proposed extension. Details are taken from each source; "not reported" does not mean absent.

Study	Setting, method, duration	AI role	Focus and relevant finding	Extension in this project
Xu et al. (2026)	Six student groups of three, one ∼45-min design task; observation, 9 interviews	Shared ChatGPT interface	Roles and negotiation; AI output became negotiable in decisions; trust and accountability set aside	Improvisation; a specific person's material; credit over repeated sessions
Suh et al. (2021) (abstract only)	15 pairs composing music with and without AI; lab study	Generative music AI	AI eased friction and altered collaborative roles	Groups; attribution of transformed offers
Trajkova et al. (2024)	24 dance students; focus groups on partnered improvisation	Proposed AI dance partner (not yet tested)	How improvisers lead, follow and value ideas	An implemented mediator rather than a partner
Mathewson and Mirowski (2017)	Live improvised theatre; development report	Chatbot performers on stage	AI accepting and making offers as a performer	AI transforming other people's offers
Epstein et al. (2020)	Online vignette experiments	AI described, not used	Agent framing shifted credit away from the artist	Credit claimed in situ by the people making the work
Caines (2026)	Research-creation study of the author's GenAI course	Web GenAI tools in teaching	Ethics, access, improvisatory skills	Contribution-level recognition in group improvisation
Proposed study (planned)	Three groups; four workshops each; episode analysis and interviews	Language-model mediator, plus a human-mediated contrast	How recognition, authority and authorship form; whether attribution and recognition diverge	—

The extension lies in the combination: following identifiable contributions through AI-mediated transformation in *repeated* collective improvisation, and asking whether technical attribution and socially negotiated recognition come apart. Contestability, mainly developed for high-stakes decisions such as hiring or sentencing (Lyons et al., 2021), offers a related vocabulary. Before the formal proposal I will extend this base into communication and cultural scholarship on collective authorship and creative labour.

3 Key terms and research questions

Three working definitions organise the study. **Recognition**: a contribution is noticed or acknowledged as meaningful. **Creative authority**: a person's interpretation or decision carries influence over what happens next. **Authorship**: participants associate ownership, credit or responsibility with the work, as felt ownership, public attribution and creative responsibility; legal ownership and copyright are outside the scope of this study. These can come apart: a contributor may be credited on screen while the group reshapes the material regardless of their view (credit without influence), and a facilitator, an interpreter or I as designer may shape the work uncredited (influence without credit). The definitions are sensitising concepts, not fixed categories. I focus on the trajectory of identifiable creative contributions while also attending to listening, facilitation, silence and refusal as forms of participation. Recognition is examined through participants' interpretations

and interactions rather than inferred solely from incorporation into the artifact. I do not assume that group decision-making is fair, that individual control is enough, or that facilitation does harm.

Central question. In collaborative digital improvisation, how do participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in the shared work?

RQ1. *Recognition.* How do participants notice, interpret and respond to differences between what they offered and how it was transformed?

RQ2. *Authority.* Through which practices, including acceptance, revision, refusal, facilitation and silence, do contributors and groups gain or lose influence over transformed material, and how do interface permissions shape those practices?

RQ3. *Authorship.* When and why do the system's technical attribution and participants' accounts of ownership, credit and responsibility diverge?

4 Theoretical framework

My primary framework is Lucy Suchman's study of human–machine reconfiguration (Suchman, 2007), which examines how agencies are figured at the interface and how humans and machines come to be treated as similar or different in practice. On this view, whether a transformed element belongs to its contributor, to the group or to the system is worked out in interaction.

Take the illustrative episode in Section 5. The system records that an amber pulse came from one participant's phrase, but the pulse also depends on the model's reading, my asset library and the approval rule. When the contributor refuses a request to brighten it, they enact an authority the interface granted, which the group can accept or contest. The question shifts from "who really made the pulse?" to how agency is distributed and attributed in that moment. Three questions follow: how participants attribute agency and authorship in talk and action ("the AI did that", "that one is mine"); how interface rules, model defaults, facilitation and my design decisions make some actions easy and others difficult; and who is held responsible when a transformation misrepresents someone. Improvisation studies adds a limited lens on listening and on how offers are taken up, ignored or refused (Caines, 2020). These concepts tell me where to look, not what I will find.

5 Research-creation methodology

The project follows the program's research-creation option, a creative work with a 150–200-page written dissertation (Joint Graduate Program in Communication & Culture, 2021). It is closest to what Chapman and Sawchuk (2012) call "research-from-creation": making the work generates material for understanding how participants engage with it.

The creative work. The creative component will comprise the shared audiovisual environment and the improvised works produced through it. Developing and revising its expressive possibilities, interaction rules and workshop scores will form part of the inquiry into how creative authority is distributed. Participants build and perform short collective improvisations, projected scenes of light, form and sound, and then discuss them. Each rule the environment embodies (who may approve, what is labelled, which assets exist) is a proposition about collective authorship that participants can test, resist or rewrite; this is how the work produces knowledge, not only data.

In an illustrative session, four improvisers receive a score: build the weather of a city none of them has visited. One types "the bus that never comes." The system proposes a slow amber pulse at the edge of the space with a low drone, with the generated interpretation "waiting as a distant repeating light." The contributor accepts it. Later another participant wants the pulse brighter; the contributor declines, and the facilitator asks the group how to proceed. Such moments of acceptance, challenge and resolution are what the study examines.

Because the creative work must be available for the committee to assess (Joint Graduate Program in Communication & Culture, 2021), it will be documented through the executable environment, workshop scores, selected consented recordings, and an annotated account of design revisions. These materials will allow the committee to examine how the creative process contributes to the dissertation's argument. A public sharing is possible but not essential.

Does the system manufacture ownership? Recording where contributions came from, and giving contributors approval rights, could encourage individual ownership where improvisers usually treat offers as shared. Do practitioners credit individual offers? Does acceptance make material available to the group, or must others ask again? Who decides when an element has become collective? Scoping interviews will investigate how practitioners already understand ownership and reuse of improvised offers. Co-design will examine whether contributor approval and visible attribution support their practices or introduce individualised ownership assumptions that participants do not share. The rules in Figure 1 are therefore provisional; one option to test is a way to "release" accepted material to the group.

Phases and provisional scale. Scoping interviews with improvisers and facilitators come first. Co-design sessions with a few practitioners then set the input form, the limits of transformation, the approval and attribution rules and accessibility provisions; these practitioners will not join the workshop groups. A provisional design would then involve three groups of three to five adult practitioners, each attending four workshops of approximately 90–120 minutes, with recruitment seeking variation in improvisational and creative-technology experience. Three groups allow comparison across compositions; four sessions give norms time to form. I would aim to include one existing ensemble and record prior relationships, since familiarity likely affects how freely people refuse one another. Each participant then has a reflection interview reviewing video of their own sessions. The final design would be refined through scoping, access negotiations and ethics review.

Arrangements. In the first workshop the group explores the environment through the manual path and writes down the procedure it will use when the group decides. Each later workshop uses one arrangement from Figure 1, in a different order for each group. The contributor-led and group-led arrangements explore different distributions of approval authority. The human-mediated arrangement keeps contributor approval and offers

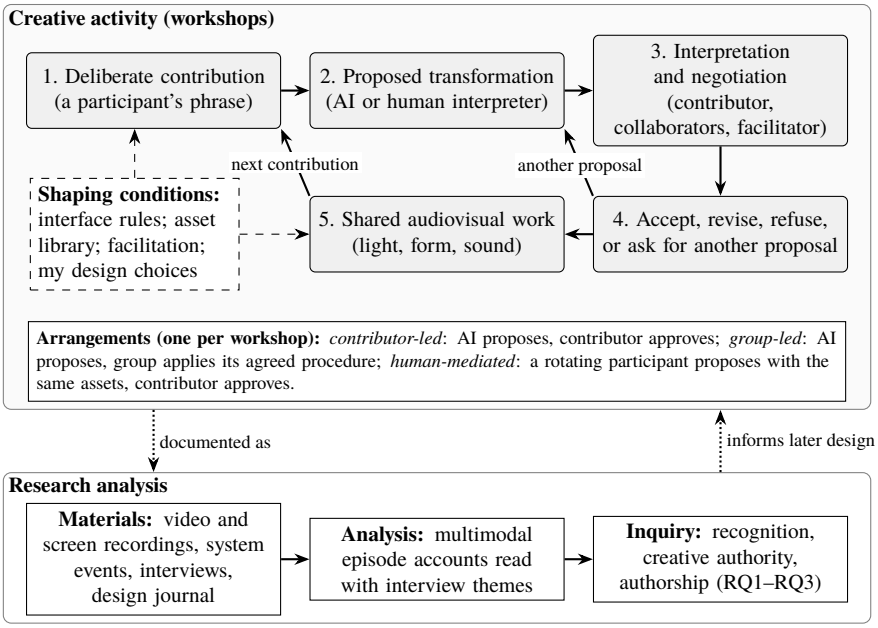


Figure 1: Proposed study (planned, not yet conducted). Solid arrows show the creative cycle in a workshop, dashed arrows show conditions that shape it, and dotted arrows connect the activity to research analysis. Recognition and authorship are interpreted from the materials; the system record does not produce them.

an additional interpretive contrast, but it also changes timing, social relationships and expectations, so the workshops support situated comparison rather than isolate a causal effect of AI. Varying the order does not remove carry-over; a group that learns to refuse in one arrangement may refuse more readily in the next, and I treat such developments as evidence of how conventions of authority form. Procedure changes and facilitator interventions are recorded in session notes, the system record and video.

6 The system and the place of AI

A participant submits a short text phrase (sketches may follow if co-design supports them). An existing language model returns one *proposal*, a choice of elements and settings from a curated library of forms, light, colour, particle "weather", sound and text, with a one-line *generated interpretation* everyone can see. That line is model output, not an explanation of why the model produced the transformation. Depending on the arrangement, contributors or groups can inspect, edit, accept, reject, request another proposal or undo, and a manual path to the same library is always open. Generative AI is included because machine-mediated reinterpretation is the object of study. The project does not presume that AI improves improvisation; it examines how participants respond to its transformations and negotiate their place in the shared work. The model does not decide whether anything enters the work, who is credited or what is removed, and makes no inferences about participants' emotions or behaviour.

Neither the model nor the library is a neutral container. Research on generative models raises concerns about stereotyped representation (Bianchi et al., 2023). Widely used language models have been shown to

reflect cultural values resembling English-speaking and Protestant European countries (Tao et al., 2024) and to favour Western-associated entities in multilingual use (Naous et al., 2024), and they still trail people in interpreting figurative phrases (Liu et al., 2022). I will therefore examine how both model interpretations and my asset choices privilege particular associations, record how metaphor, idiom and non-English phrases are read, and discuss misreadings with their authors. Conclusions will concern the people in the study, not cultures or communities it does not include.

Feasibility. The technical design is deliberately modest, so that effort goes into the expressive and social rules under study. Unity applies only permitted actions with bounded settings; model output is checked against them and never run as code. I will first test a local open-weight model to avoid third-party processing, otherwise an institutionally approved hosted service receiving only the phrase, without training or fine-tuning any model.

7 Case-study context and access

Subject to discussion with Dr. Caines, these workshops could examine how improvisational practices help participants question and reshape the authority built into creative technologies, a question close to the title of her SSHRC Insight project, *Interfacing, Playtesting, Resetting: Reframing Technology Use Through Improvised Creative Technologies* (Goodfellow Craig, 2026). The particular setting and relationship to her funded project would be determined collaboratively. Only its title and funding are public, and I do not assume it includes this study, these participants or any AI component.

Otherwise I would use open calls to Toronto improvisation, theatre and sound communities and to AMPD students, excluding anyone I teach. As in Chalk's (2022) study of Actual Play producers, recruitment would be iterative: after the first group I would check whose practices are missing, adjust later calls, and report remaining gaps. I will not work with disability-focused or Indigenous communities without established partnerships, community governance and relevant expertise.

8 Analysis

I will construct multimodal accounts of selected episodes, linking video of participants' words and actions with screen recordings and the system's log. An episode runs from a contribution, through its transformation and the group's response, to the next creative action. Episode selection will include routine acceptance and non-response as well as explicit disagreement. I will document selection decisions and compare episodes across groups, arrangements and stages of collaboration, including cases that challenge emerging interpretations.

The episode analysis adapts the video microethnography McArthur (2015) used to study players' interactions with interfaces. Reflection interviews are analysed through reflexive thematic analysis (Braun and Clarke, 2019); their themes show how participants describe ownership and authority in their own terms, contextu-

alising an episode without settling what it meant. System records show what happened in the software, not a "true" account of intention against which participants are judged. Discrepant cases, such as a group that ignores the attribution display entirely, either revise an emerging explanation or limit its scope. In keeping with a reflexive approach, I will keep an analytic trail rather than seek inter-rater agreement, and the design journal records how my choices as designer and facilitator shape what participants can do.

9 Ethics, consent and accessibility

Consent will be specific, ongoing and revisable (Im et al., 2021; Strengers et al., 2021), given separately for recording, AI processing, use of recordings in the examination, and public dissemination. Provenance will support identifying and, where feasible, removing attributable contributions. Participants will agree on withdrawal procedures and deadlines before the workshops, including the limits of removing contributions from jointly developed or publicly disseminated works. Credit is agreed with participants, whose time is paid; data are minimised, pseudonymised and kept for stated periods. Full participation without AI is a design commitment the workshops will test, through reduced motion, resizable text, alternatives to sound-only information, facilitator-operated controls and the manual path. Participation alone does not make a project equitable (Birhane et al., 2022).

10 Intended contribution

The project investigates whether technical attribution and social recognition can diverge in collective creative work. A provenance log may record who supplied a phrase while collaborators treat the transformed material as belonging to another person, to the group, or to nobody in particular. The study will examine how contributors gain or lose authority over transformed material, and how interface permissions, facilitation and group conventions shape that process. Its intended contribution is an explanation of these relationships, rather than a general claim that more individual control produces fairer collaboration.

This is a proposition to investigate, not a finding. RQ3 asks whether and when divergence occurs; RQ1 and RQ2 trace the recognition and authority that would produce it. If divergence appears, the account would reach beyond these workshops, to creative platforms that rely on attribution records and the improvised conventions they meet; if not, the study would explain why the two stayed aligned. If AI proves unnecessary in some arrangements, the contribution would lie in explaining when, how and why, and what that reveals about collective creativity.

11 Relationship to my master's research

My master's research at Simon Fraser University asks how interfaces make a person's state perceptible to others, through label-free heart-rate cues for remote teammates in a cooperative Unity game and a Unity

environment that responds to the player's heart rate. Both manuscripts are under revision following peer review (Jahandideh, 2026a,b). In the teammate study, participants tied their willingness to use the cue to control over it, asking for opt-in sharing, an immediate off switch and separate permissions (Jahandideh, 2026b). As the designer, I also noticed that the classification behind the display still sorted people into evaluative categories they never saw. Both lead to this PhD's question, who controls how a shared system interprets people, now asked of deliberate creative contributions rather than bodily signals. My path from architecture and landscape architecture to building and evaluating networked systems in HCI prepares me to design a shared environment as a creative work and to study how people interpret it.

12 Provisional chapter structure

(1) Introduction; (2) literature and framework: co-creation, credit, contestation and human–machine reconfiguration; (3) methodology: research-creation, workshops and reflexivity; (4) the environment as a creative work and the ownership its design assumes; (5) recognition and authority: how contributions are noticed, transformed, taken up or refused; (6) authorship: attribution, recognition and responsibility, and where they diverge; (7) conclusion.

13 Provisional timeline (full-time)

Year 1 (2027–28)	Courses (CMCT 7000, 7200, 7005 in the current degree requirements, and electives; offerings may change); literature review; ethics preparation; tests with a local model.
Year 2 (2028–29)	Comprehensive examinations; formal proposal (maximum 3,500 words) with the supervisory committee (Joint Graduate Program in Communication & Culture, 2016); ethics approval; scoping interviews.
Year 3 (2029–30)	Co-design; workshops with three groups; reflection interviews; analysis.
Year 4 (2030–31)	Analysis; documentation of the creative work for examination; writing and defence. Year 5 is held in reserve.

14 Program and supervisor fit

The project belongs to Technology in Practice, with Media & Culture as a minor. Dr. Caines's work on improvisation as a research methodology (Caines, 2020) and socially responsive creative technologies (Caines and Watson, 2023) speaks to its setting and methods, and her research-creation study of generative AI in classrooms attends to ethics, access, collaboration and improvisatory skills (Caines, 2026). I carry that concern to contribution-level recognition and contested authorship in group improvisation.

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